Limit behavior of Polya urn schemes

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Abstract

The long term behavior of the composition of Polya's urn schemes is analyzed by means of simple martingale arguments. The results hold even under the assumption that random numbers of balls are added to the urn depending only on the color of the ball obtained in the successive extractions.

1 Polya's Urn schemes

Urn schemes, devised by George Polya, are a useful tool to think about probability beyond the scope of independence, too predominant in introductory courses. They help to illustrate many essential concepts such as *exchangeability*, *martingales*, *stochastic convergences* and *limit laws*.

An urn scheme considers an urn with balls of two different colors: $\mathcal{A}$mber and $\mathcal{B}$lue. Successive extractions are performed at random and the obtained ball is returned to the urn together with a certain number of additional balls of each color. Initially there is a total of t balls in the urn, α of which are amber and $\beta = t - \alpha$ are blue. The urn scheme is characterized by the parameters:

$$[\ \underbrace{a'\ ,\ \ a}_{amber}\ ;\ \underbrace{b\ \ ,\ \ b'}_{blue}\ ]$$

where a' and a stand for the number of amber and blue balls respectively added when an amber ball is obtained, while b amber and b' blue balls are adjoined if a blue ball is extracted. Negative values for some of these values are usually admitted and to add $-x$ balls means to remove x balls of the urn. However, in this case it may happen that the balls of some color are exhausted and the extractions cannot be pursued according to the rules of the model.

The most practical assumption is that the total number of added balls is fixed, say $c > 0$; so that $a' = c - a$ and $b' = c - b$. In this case the total number of balls in the urn is not random, but independent of the successive colors previously obtained. Unless otherwise stated, this will be the considered model:

$$[c - a,\ a; b,\ c - b], \qquad c > 0.$$

We will denote

- Y_k the color of the ball obtained in the trial $k \geq 1$: $Y_k = 1$ if it is amber, $Y_k = 0$ if it is blue.
- $X_n = \sum_{k=1}^{n} Y_k$, the total number of amber balls obtained in the first n extractions.

One can allow the possibility that a and b could have random values. To this end, assume that $\{a_k\}_{k\geq 1}$ and $\{b_k\}_{k\geq 1}$ are two sequences of independent identically distributed random variables with integer values, independent also of $\{Y_k\}_{k\geq 1}$. Then if $Y_k = 1$, a_k blue balls and $c - a_k$ amber balls are added to the urn, while if $Y_k = 0$ the blue ball is returned to the

urn together with b_k amber balls and $c - b_k$ blue balls. Specifically a_k and b_k would have a finite range of values (for instance $0 \le a_k, b_k \le c$) and $\bar{a}$ and $\bar{b}$ will denote the means of a_k and b_k respectively.
The references [2], [3] and [4] collect the known results about this subject. [6] also contains a review of the main papers in the area.
The aim of this paper is to show that the long term frequencies of each color will always be stabilized around some value, fixed or random, and therefore it can never oscillate indefinitely. This conclusion is established in Section 3. As an introduction, Section 2 presents some well known results to be used later.

2 The Polya–Eggenberger model

The most complete analysis of an urn scheme corresponds to the Polya–Eggenberger model for which $a = b = 0$. Explicitely, when a ball of any color is obtained, it returns to the urn only with c new balls of the same color. In these conditions, for any $(i_1, i_2, \ldots, i_n) \in \{0,1\}^n$, it will be

$$\mathbb{P}\{Y_1 = i_1, \ldots, Y_n = i_n\} = \frac{\alpha\,(\alpha + c) \cdots (\alpha + (r-1)c)\;\beta(\beta + c) \cdots (\beta + (s-1)c)}{t\,(t + c) \cdots (t + (n-1)c)} \tag{2.1}$$

where $r = \sum_{j=1}^{n} i_j$ is the number of ones and $s = n - r$ the number of zeros in $(i_1, \ldots, i_n)$. In fact, the successive factors in the denominator give the number of balls in the urn when the successive extractions are done. If i_j is the first index with value 1, at the time of extraction j, there are α amber balls in the urn; next $\alpha + c$ amber balls are in the urn when the second amber ball is obtained; and so on. Similarly, $\beta, \beta + c, \beta + 2c, \ldots$ are the number of blue balls in the urn when the successive blue balls are obtained. An obvious modification is needed for $i_1 = \cdots = i_n$, since no α (or β) factors appear if all them are zeros (or ones). Upon dividing by c^n the numerator and denominator, we get

$$\mathbb{P}\{Y_1 = i_1, \ldots, Y_n = i_n\} = \frac{\Gamma(\alpha/c + r)}{\Gamma(\alpha/c)} \frac{\Gamma(\beta/c + s)}{\Gamma(\beta/c)} \frac{\Gamma(t/c)}{\Gamma(t/c + n)} = \frac{\beta(\alpha/c + r, \beta/c + s)}{\beta(\alpha/c, \beta/c)} \tag{2.2}$$

As these probabilities depend on $(i_1, i_2, \ldots, i_n)$ only through n and r (and $s = n - r$), any permutation of $(i_{\pi(1)}, \ldots, i_{\pi(n)})$ has the same probability or, equivalently,

$$\mathbb{P}\{Y_{\pi(1)} = i_1, \ldots, Y_{\pi(n)} = i_n\} = \mathbb{P}\{Y_1 = i_1, \ldots, Y_n = i_n\}$$

so that $\{Y_k\}_{k \ge 1}$ is an exchangeable sequence of random variables. We can then use the well known result:

Theorem 1 (de Finetti) *Let $\{Y_k\}_{k \ge 1}$ be an exchangeable sequence of random variables with values in $\{0,1\}$. There exists a random variable U, with distribution θ on $(0,1)$, such that, conditionally on U, $\{Y_k\}_{k \ge 1}$ are independent random variables with distribution: $\mathbb{P}\{Y_k = 1 | U\} = U$.*
Therefore, if $i_1 + \cdots + i_n = r$, it is

$$\mathbb{P}\{Y_1 = i_1, \ldots, Y_n = i_n\} = \int_0^1 u^r (1-u)^{n-r}\, \theta(du) \tag{2.3}$$

and moreover $\frac{1}{n} \sum_{k=1}^{n} Y_k \longrightarrow U$ a.s.

The last sentence states that the ratio X_n/n of amber balls obtained in the first n attempts converges, with probability 1, to a random value U that may have a different value each

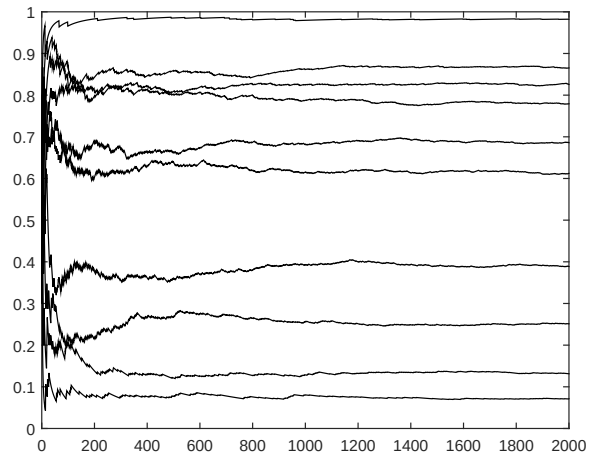

Figure 1: Proportion of amber balls when $a = b = 0$

time the experience is performed. But long-term oscillations will never be observed in the average X_n/n nor on the proportion of amber balls in the urn:

$$p_n = \frac{\alpha + cX_n}{t + cn} \sim \frac{X_n}{n}.$$

A simple MatLab program allows to draw the graphic of Figure 1, whose curves (n, p_n) correspond to 10 realizations of 2000 extractions from an urn, with initially 3 amber balls and 2 blue balls, to which 5 balls of the obtained color are added. This is the same kind of phenomenon observed when extracting balls from an urn with fixed composition, except for the key distinctness that the limit is not a preestablished value, but a different value of U in each replay.

Concerning the distribution θ with which the value of U is chosen, according to (2.3) it must be

$$\int_0^1 u^r (1-u)^{n-r} \theta(du) = \frac{\beta(\alpha/c + r, \beta/c + s)}{\beta(\alpha/c, \beta/c)}\,. \tag{2.4}$$

But, when θ is the *beta* distribution with parameters α/c and β/c, the left member is

$$\int_0^1 u^{\alpha/c+r-1}(1-u)^{\beta/c+n-r-1} \frac{du}{\beta(\alpha/c, \beta/c)} = \frac{\beta(\alpha/c + r, \beta/c + n - r)}{\beta(\alpha/c, \beta/c)}$$

and the equality (2.3) holds. Obviously, there is no other distribution with the same property, since (2.4) determines, for $r = n$, all the moments of θ. Shortly, if the urn composition is chosen with distribution $beta(\alpha/c, \beta/c)$, by replacing only the extracted ball, the sequence $\{Y_k\}$ will have the same distribution than in the Polya-Eggenberger model.

These results are quoted without proof in [2] and were implicit in the original paper of Eggenberger and Polya. They can also be found in [5, Examples VII, 4(a) and 9(a)].

3 A martingale analysis

We now return to the general setting of the first section, with c balls added after each extraction k, of which a_k are blue if the extracted ball is amber and b_k are amber if the extracted ball is blue. We will analyze some kind of almost sure asymptotic behavior of the urn composition, based only on the means $\bar{a}$ and $\bar{b}$.

Let $\mathcal{F}_n = \sigma(\{Y_k, a_k, b_k\}_{1\le k\le n})$ be the σ-field of events depending on the results of the first n moves in a stochastic Polya urn scheme. The fraction of amber balls in the urn after the first n moves:

$$p_n = \frac{\alpha + \sum_{k=1}^n Y_k(c-a_k) + \sum_{k=1}^n (1-Y_k)b_k}{t+cn} \tag{3.5}$$

is $\mathcal{F}_n$-measurable. Further

$$\mathbb{E}[p_{n+1}|\mathcal{F}_n, a_{n+1}, b_{n+1}] = \frac{p_n(t+cn) + p_n(c-a_{n+1}) + (1-p_n)b_{n+1}}{t+c(n+1)}$$

and

$$\mathbb{E}[p_{n+1}|\mathcal{F}_n] = p_n + \frac{(1-p_n)\bar{b} - p_n\bar{a}}{t+c(n+1)}. \tag{3.6}$$

More precisely, p_n gives the probability of obtaining an amber ball until the stopping time $\tau = \inf\{n \ge 1 | p_n \notin [0,1]\}$. When $\tau < \infty$ and $p_\tau > 1$, the experiment should end due to the lack of blue balls; similarly, for $\tau < \infty$ and $p_\tau < 0$, extractions are stopped due to the lack of amber balls. Obviously if $0 \le a_k, b_k \le c$, the numbers of balls of both colors are increasing and $\tau = \infty$.

Proposition 1 *Assume that $\bar{a} \le 0 < \bar{b}$.*
(i) As long as $p_n \in [0,1]$, $\{p_n\}$ is a submartingale.
(ii) With probability 1, $\tau < \infty$ or $p_n \to p_\infty \in [0,1]$ (that can be a random variable). Moreover $p_n \xrightarrow{L^1} p_\infty$ conditional on $\{\tau = \infty\}$.
(iii) If $\bar{a} = 0$, it is $p_\infty = 1$. For $\bar{a} < 0$, $\mathbb{P}\{\tau < \infty\} = 1$.

Proof: (i) $\mathbb{E}[p_{n+1}|\mathcal{F}_n] \ge p_n$ if and only if $p_n(\bar{a}+\bar{b}) \le \bar{b}$. And, indeed, $p_n(\bar{a}+\bar{b}) \le p_n\bar{b} \le \bar{b}$.
(ii) Conditional on $\{\tau = \infty\}$, p_n is a bounded submartingale, convergent to a random variable p_∞ almost surely and in L^1.
(iii) When $\bar{a} = 0$, let $\bar{p}_n = \mathbb{E}[p_n|\tau = \infty]$ which increases to $\bar{p}_\infty = \mathbb{E}[p_\infty|\tau = \infty]$. Assuming that some trajectory in $\{\tau = \infty\}$ satisfies $1 - \bar{p}_\infty > \varepsilon$ for some $\varepsilon > 0$, it would verify

$$\bar{p}_{n+1} > \bar{p}_n + \frac{\bar{b}\,\varepsilon}{t+c(n+1)}.$$

This gives a contradiction, because the last terms form a divergent series.
For $\bar{a} < 0$, if $\bar{b}/(\bar{a}+\bar{b}) = \delta > 1$, assume that $\mathbb{P}\{\tau = \infty\} > 0$ and then $\bar{p}_n \le 1$ for all the trajectories in $\{\tau = \infty\}$ and all n. But

$$\bar{p}_{n+1} \ge \bar{p}_n + (\bar{a}+\bar{b})\frac{\delta - 1}{t+c(n+1)}$$

will give $\bar{p}_\infty = \infty$. A similar argument holds if $\bar{b}/(\bar{a}+\bar{b}) = \delta < 0$ and it is even simpler for $\bar{a}+\bar{b} = 0$. $\square$

Shortly, for $\bar{a} < 0 < \bar{b}$ some color exhausts in a finite time, while for $\bar{a} = 0$ the urn is asymptotically depleted of blue balls, if any color is not exhausted in a finite time. In this last case, $p_n \to 1$ and therefore:

Corollary 1 *When $\bar{a} = 0 < \bar{b}$, in $\{\tau = \infty\}$ it is $X_n/n \longrightarrow 1$ almost surely.*

Proof: Let us consider $Z_k = Y_k - p_{k-1}$, for which $\mathbb{E}[Z_k|\mathcal{F}_{k-1}] = 0$. Moreover, since $\mathbb{E}[Y_k|\mathcal{F}_{k-1}] = p_{k-1}$, it is

$$\mathbb{E}[Z_k^2] = \mathbb{E}[Y_k] - 2\mathbb{E}[Y_k p_{k-1}] + \mathbb{E}[p_{k-1}^2] = \mathbb{E}[p_{k-1}] - \mathbb{E}[p_{k-1}^2] \le \frac{1}{4}$$

and therefore $\sum_{k=1}^{\infty} k^{-2}\mathbb{E}[Z_k^2] < \infty$. Thus, the strong law of large numbers for martingales given in [5, VII, Theorem 3] asserts that, almost surely,

$$\frac{1}{n}\sum_{k=1}^{n} Z_k = \frac{X_n}{n} - \frac{1}{n}\sum_{k=1}^{n} p_k \longrightarrow 0 \qquad \text{or} \qquad \frac{X_n}{n} \longrightarrow 1$$

since $p_n \to 1$ in $\{\tau = \infty\}$. $\square$

A simpler argument may be based on the redraft of (3.5):

$$\Big(\frac{t}{n}+c\Big)p_n = \frac{\alpha}{n} + c\,\frac{X_n}{n} - \frac{X_n}{n}\,\frac{1}{X_n}\sum_{k=1}^{n} Y_k a_k + \frac{1}{n}\sum_{k=1}^{n} b_k - \frac{X_n}{n}\,\frac{1}{X_n}\sum_{k=1}^{n} Y_k b_k. \tag{3.7}$$

The Borel–Cantelli lemma in [1, Theorem 5.3.2] gives $\{X_n \to \infty\} = \{\sum_{n=1}^{\infty} p_n = \infty\}$, so that $X_n \to \infty$ almost surely in $\{\tau = \infty\}$. Thus $(1/X_n)\sum_{k=1}^{n} Y_k a_k \to \bar{a} = 0$ and $(1/X_n)\sum_{k=1}^{n} Y_k b_k \to \bar{b}$ whatever values the sequence $\{Y_k\}_{k\geq 1}$ may have. Therefore it follows $c = \bar{b} + (c - \bar{b})\lim_n X_n/n$ and $\lim_n X_n/n = 1$, under the unavoidable requirement $\bar{b} \neq c$.

Of course, for $\bar{b} \leq 0 < \bar{a}$, the possible outcomes are reversed. While $\{p_n\}$ is a supermartingale, the proportion of blue balls is a submartingale. Then, $\tau < \infty$ for $\bar{b} < 0$ and, if $\bar{b} = 0$, either $\tau < \infty$ or $p_n \to 0$.

We now turn to the case where $\bar{a}, \bar{b} \neq 0$ have the same sign or, equivalently,

$$p^\star = \frac{\bar{b}}{\bar{a} + \bar{b}} \in (0, 1).$$

Proposition 2 *Assume that $\bar{a}, \bar{b} > 0$.*
(i) As long as $p_n \in [0, p^\star]$, $\{p_n\}$ is a submartingale while, when $p_n \in [p^\star, 1]$, $\{p_n\}$ is a supermartingale.
(ii) With probability 1, $\tau < \infty$ or $p_n \to p^\star$.

Proof: (i) If $p_n \in [0, p^\star]$ the second term in the right hand side of (3.6) is nonnegative; thus $\{p_n\}$ is a submartingale. On the contrary, for $p_n \in [p^\star, 1]$, the indicated term is nonpositive and $\{p_n\}$ is a supermartingale.
(ii) Within the event $\{\tau = \infty\}$ let $\bar{p} = \limsup_n p_n$ and $\underline{p} = \liminf_n p_n$ and let us consider $C = \{\tau = \infty, \underline{p} < \bar{p}\}$. The event $C \cap \{\underline{p} > p^\star\}$ has probability zero, because those trajectories are, from some n onwards, trajectories of a submartingale with two different cluster points. For the same reason, the probability of $C \cap \{\bar{p} < p^\star\}$ is also zero. Hence, $C \cap \{\underline{p} < p^\star < \bar{p}\}$ differs from C in a set of probability zero.
Now, let $C_\delta = C \cap \{\underline{p} < p^\star < p^\star + \delta < \bar{p}\}$. The trajectories in C_δ must perform an infinite number of upcrossings of the interval $(p^\star, p^\star + \delta)$ through positive steps of size

$$p_{n+1} - p_n = \begin{cases} \dfrac{c - a_{n+1} - cp_n}{t + c(n+1)} & \text{if } Y_{n+1} = 1 \\[2ex] \dfrac{b_{n+1} - cp_n}{t + c(n+1)} & \text{if } Y_{n+1} = 0 \end{cases}$$

that is less than any $\varepsilon > 0$ for n large enough. Therefore the probability of C_δ is bounded by $[(p^\star + \delta) \vee (1 - p^\star)]^{\delta/\varepsilon}$ and, this being true for any $\varepsilon > 0$, it should be $\mathbb{P}(C_\delta) = 0$. A similar reasoning shows that $C'_\delta = C \cap \{\underline{p} < p^\star - \delta < p^\star < \bar{p}\}$ has also probability zero and consequently $\{\tau < \infty\} \cup \{\underline{p} = p^\star = \bar{p}\}$ has probability one. $\square$

Repeating the proof of Corollary 1, the convergence $X_n/n \to p^\star$ follows from $p_n \to p^\star$.

In [4, Corollary 3.2] the assertion $p_n \xrightarrow{P} p^\star$ is attributed to Bagchi–Pal (1985), when $a_k = a$ and $b_k = b$ are non random values, under the additional condition $a + b \geq c/2$. Later [4,

Theorem 6.2] reproduces a similar result from Smythe (1996) which again only asserts the convergence in probability for nonrandom parameters. For the Friedman (1949) model, in which $a = b > 0$ is a fixed non random integer, the almost sure convergence $p_n \to 1/2$ was established by Freedman (1965). A different proof, due to Ornstein, can be found in [1, Example 5.4.5].

For $\bar{a}, \bar{b} < 0$ a result similar to Proposition 2 holds:

Proposition 3 *Assume that $\bar{a}, \bar{b} < 0$.*
(i) As long as $p_n \in [0, p^\star]$, $\{p_n\}$ is a supermartingale while, when $p_n \in [p^\star, 1]$, $\{p_n\}$ is a submartingale.
(ii) With probability 1, $\tau < \infty$ or $p_n \to p^\star$.

Assertion *(i)* is obvious and the reasoning of *(ii)* in Proposition 2 needs no change to hold in this case. However the path of $\{p_n\}$ is now much more unstable since, as opposed to the previous case, there is here a trend upwards above $p^\star$ and downwards below $p^\star$. Hence the probability of $\{\tau < \infty\}$ is greater for $-\bar{a}, -\bar{b}$ than for $\bar{a}, \bar{b} > 0$.

Finally, we may consider the state of affairs for $\bar{a} = \bar{b} = 0$. Under such circumstances (3.6) shows that $\{p_n\}$ is a bounded martingale while $p_n \in [0, 1]$. Therefore $p_n \to p_\infty$ in $\{\tau = \infty\}$. But so, nothing can be said about p_∞. Fortunately, one can show

Lemma 1 *Assume that $\bar{a} = \bar{b} = 0$.*
As long as $p_n \in [0, 1]$, (2.1) holds. Consequently, $\{Y_k\}_{k \geq 1}$ is an exchangeable sequence of random variables conditional on $\{\tau = \infty\}$.

Proof: If $p_n \in [0, 1]$, it is

$$\mathbb{P}\{Y_{n+1} = 1 | \mathcal{F}_n\} = p_n = \frac{\alpha + \sum_{k=1}^n Y_k(c - a_k) + \sum_{k=1}^n (1 - Y_k) b_k}{t + cn}$$

thus, since $\bar{a} = \bar{b} = 0$, we have

$$\mathbb{P}\{Y_{n+1} = 1 | Y_1, \ldots, Y_n\} = \frac{\alpha + c \sum_{k=1}^n Y_k}{t + cn} = \frac{\alpha + cr}{t + cn}$$

in $\{X_n = r\}$ and

$$\mathbb{P}\{Y_{n+1} = 0 | Y_1, \ldots, Y_n\} = \frac{\beta + c \sum_{k=1}^n (1 - Y_k)}{t + cn} = \frac{\beta + sc}{t + cn}$$

in $\{X_n = n - s\}$. Hence, (2.1) holds and $\{Y_k\}$ is exchangeable when $\{\tau = \infty\}$. $\square$

According to the previous result and using again de Finetti's theorem:

Proposition 4 *Assume that $\bar{a} = \bar{b} = 0$.*
Conditional on $\{\tau = \infty\}$, the limit $p_\infty = \lim_n p_n$ exists almost surely and has the same beta$(\alpha/c, \beta/c)$ *distribution than in the Polya–Eggenberger model.*

4 Final comments

Our main conclusion is that a Polya urn scheme $[c - a, a; b, c - b]$, even with random values of a and b, can never have long term oscillations. It either fails for lack of some kind of balls or, on the contrary, the proportion of amber balls converges to the fixed value $p^\star = \bar{b}/(\bar{a} + \bar{b})$ or, if $\bar{a} = \bar{b} = 0$, to a random value p_∞ with distribution *beta* $(\alpha/c, \beta/c)$. The scarce role of c is quite unexpected.

It is usual in the literature about Polya's urn schemes to consider only *tenable* models that *withstand the test of time... and we can perpetuate the drawing according to the given scheme*

on every possible stochastic path [4, Section 3.1]. In our terms that means $\mathbb{P}\{\tau = \infty\} = 1$. In the indicated reference, conditions for tenability about non random values of $a', a, b, b', \alpha, \beta$ are exposed. But such approach does not fit in the case of random parameters (unless the analysis is restricted to the case $0 \leq a_k, b_k \leq c$).

It would be desirable to complete the results exposed here with a research about the value of $\mathbb{P}\{\tau = \infty\}$ as a function of the parameters α, β and c and the distributions of a_k, b_k. Although this issue seems unsuccessful from a theoretical point of vue, it may be analyzed by means of simulations for any fixed conditions. For instance, for the zero mean distributions $a_k = [-5, -2, 2, 5]$, $b_k = [-4, 0, 4]$, each one with equal probabilities, $\alpha = \beta = 10$ and $c = 3$, every replay of 5000 steps of the scheme gives place to a curve as in Figure 1, although now it can overflow from $[0, 1]$ at a time τ. In 100 replays of this scheme the observed values of τ were:

τ	> 5000	4	5	6	7	8	9	10	12	14	15	16	19	20	21	22	23
%	42	4	5	1	4	4	3	1	3	2	3	1	2	2	1	1	3
τ		25	26	27	29	36	37	44	57	85	87	90	99	101	186	492	826
%		3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

This is a strong clue that $\mathbb{P}\{\tau = \infty\} \simeq 0.42$. Of course this probability is very sensitive to changes in α, β and c.

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